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# A role-playing game to stimulate thinking about vineyard management practices to limit pesticide use and impacts

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ABSTRACT

Pesticides harm the environment and the health of applicators and rural populations. Although several levers exist to help reduce agricultural pesticide use, farmers hesitate to use them because they can require specific training, costly materials, and additional worktime while often decreasing yields. In this study, our objective was to design and develop an ad hoc role-playing game that can be used to design strategies of change for agricultural practices to reduce the unwanted impacts of pesticide use at farm and watershed scale. The main challenge was to create a game that is realistic and relevant to actors while bringing new knowledge to foster learning. We developed a participatory process to build the game and then complemented it with original contributions from research: the results of innovative field experiments and the design of simple assessment indicators corresponding to the evaluation criteria of actors. Our case study was a vineyard watershed subject to heavy pesticide use. Interview surveys with farmers and workshops with local actors (including farmers) and experts were employed to build the game, which included a spatial typology of farms, a simple calculator of potential impacts, and two types of roles. In the role of winegrower, the participants propose changes in practices for a virtual farm. In the role of public administrator, they evaluate the winegrowers' requests and the evolution of potential impacts, and then suggest further changes. The game was played with local actors and experts, separately. Game results were analyzed in terms of changes in practices, subsidy requests, and the evolution of potential impacts, both at the scale of farm types and the entire watershed.In the two game sessions, all participants modified the practices of their virtual farms. This revealed common and divergent choices between the group of experts and the group of local actors. Experts requested more plot-scale subsidies. They were also more ambitious in managing fungi and decreasing glyphosate use. At the watershed scale there were few differences between the potential impacts of the strategies as designed by the two different groups. However, the different attitudes of local actors and experts revealed different perceptions on investment, risks, and subsidies.This is the first game developed at a territorial scale to co-design and assess agricultural systems aimed at reducing unwanted impacts of pesticide use. The game is limited by the absence of economic and yield impacts, which should be a priority in the development of future versions of the game.

## 1. Introduction

Pesticides are used extensively in modern agriculture (Stoate et al., 2001), and this use is known to be harmful to the environment and human health (Enserink et al., 2013). Health threats have been noted particularly for pesticide applicators and rural populations (Alavanja et al., 2004; Elbaz et al., 2009). The gravity of the threat depends on the level and length of exposure and the type of active substance (e.g., carcinogenic). Threats to ecosystem health involve the soil, water, and air, in effect, all aspects of the environment which are polluted by direct pesticide application or by secondary contamination via atmospheric or hydrological pathways. Pesticides have been shown to increase earthworm mortality (Pelosi et al., 2014), lead to fish mortality (Murthy et al., 2013), and cause pollinator decline (Brittain et al., 2010). Water quality (both surface and underground, including drinking water catchments) is also threatened by pesticides, especially herbicides (Louchart et al., 2001), leading to many situations where water quality standards are not met. For example, 35% of monitored points in French groundwaters do not meet drinking water standards (Ministère de la Transition Ecologique, 2020).

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Reductions in pollution from pesticides is an objective of agricultural policies in several European countries (Barzman and Dachbrodt-Saaydeh, 2011). Limiting the use of pesticides and limiting the environmental impact associated with their use are the two main options for reducing this pollution. The numerous levers employed to achieve these decreases usually involve buffer zones (including agro-ecological infrastructures), agricultural management and practices, and the spatial distribution of pesticide use in the landscape. For example, hedgerow infrastructures can intercept spray drift from pesticide-treated fields (Lazzaro et al., 2008), and specific management of ditches for collecting field runoff water can increase herbicide retention and limit pesticide contamination of surface waters (Dollinger et al., 2017). In terms of agricultural practices, the main levers depend on the crop and the target (weeds, fungi, insects). Decreasing herbicide use in weed management is predominantly accomplished by two means: mechanical destruction (tillage and plowing), and grassing (Anses, 2020a, 2020b). However, these options can be hampered by soil type, slope, and the type of weeds; they also imply additional worktime and costs and generate risks for yield (Anses, 2020a, 2020b). Alternative methods for controlling insects include observations and the use of biocontrol agents, such as predator release strategies and pheromone-based products for mating disruption (Pertot et al., 2017). Biocontrol products (e.g., biopesticides based on natural molecules) can also be used to reduce the application of fungicides (e.g., Pertot et al., 2017). Reductions in fungicide use can also be accomplished with two other important levers: disease-resistant cultivars, and decision tools to optimize (Pertot et al., 2017). The latter can make it possible to delay the 1st treatment, thus decreasing the quantity of fungicides required (e.g., Mailly et al., 2017). Still another lever involves the devices used for spraying. Drift-reducing sprayers, such as air-blast sprayers, can help decrease the quantity of pesticides applied by improving the precision and efficiency of delivery to the canopy (Pergher and Petris, 2008; van de Zande et al., 2008). Optimal solutions often combine levers that are temporally and spatially coordinated. However, they all face obstacles that constrain adoption by farmers. For example, biocontrol agents suffer from variable effectiveness linked to climatic and site-specific factors, application techniques, and costs (Lamichhane et al., 2017).

Several methods have been proposed to design strategies for practices aimed at reducing the use and/or impacts of pesticides. The primary decision-makers for agricultural production are the local stakeholders (Primdahl, 1999); they can propose solutions that may be better adapted to local contexts (Reed, 2008). Among participatory methods for strategy design, those combining design and modelling (i.e., participatory modelling) can facilitate decision-making tailored to the mitigation of specific issues (e.g., Barnaud et al., 2008). Co-Click'eau, for example, is a participatory method aimed at developing catchment action plans (land use and production type scenarios), whose performances are then simulated using simple indicators of pesticide use (not impact) and aggregation tools (Chantre et al., 2016). Another type of approach, which has recently gained importance, is based on serious games involving computer simulations. This approach helps to stimulate learning while taking account of stakeholders' viewpoints and their decision-making context (Becu et al., 2008). Recent studies have proposed serious games that stimulate learning about agro-ecological practices and their impacts on environmental (including pesticide use), economic, and social sustainability at the farm scale (SEGAE, Jouan et al., 2021). Another is aimed at designing more environmental-friendly technical management plans for vineyards at the field scale, and combining them with life cycle assessments, including the toxicity of pesticides in freshwater and humans (VitiPoly, Rouault et al., 2020). These two studies addressed limited spatial scales and examined the impacts of pesticides in a simplified (scoring in SEGAE) or partial (only humans and water) manner. At a territorial scale, Della Rossa et al. (2022) developed and employed a role-playing game to co-design agricultural systems to reduce herbicide pollution, which were assessed with a process-based model simulation. This study however relied on pre-existing and previously parameterized modelling tools, and it focused on herbicides only, so it did not consider the whole system of practices and impacts. There is currently a research gap in methods to co-design and assess agricultural systems aimed at reducing the uses and impacts of all pesticides at territorial scale.

In this study, our objective was to (1) design and develop an *ad hoc* role-playing game associated with a simple calculator of means-based and results-based indicators (hereafter named CAPPIP: Changes in Agricultural Practices and Impacts of Phytosanitary Products) and (2) play the game to design strategies for changes in agricultural practices to reduce the impacts of pesticides at farm and watershed scale. We chose a vineyard watershed for our case study because grapevine cultivation involves significant pesticide use. The French national Treatment Frequency Index (TFI) provides a measurement of the intensity of pesticide use (Pingault et al., 2009). The national mean TFI for grapevine cultivation was 12.4 in 2019 (Agreste, 2021). By comparison, the mean TFI for cereals (barley, wheat) ranged between 4.2 and 5.1 in 2017 (Agreste, 2020). The main challenges were to build and employ a game that would be playable and relevant for actors while being realistic (current situations and options for change) and bringing new knowledge to foster discussions and learning on new agricultural strategies and their potential effects on pesticide use and its impacts on the environment and humans. To overcome these challenges, we created a participatory process to build the game and then complemented it with original contributions from research. These included results of innovative field experiments and the design of simple assessment indicators corresponding to the evaluation criteria of actors.

## 2. Material & methods

### 2.1. Method overview

The process of designing and assessing changes in wine-growing practices was carried out in a series of steps organized around the development and enactment of a serious game (Fig. 1). Surveys and workshops were employed in four steps: (1) characterize the wine-growing area; (2) identify strategy objectives, the indicators for their assessment, and the levers of change; (3) gather information to build the game; and (4) play the game to design and assess changes in practices.

The surveys and workshops were conducted with local stakeholders (winegrowers), local actors, and experts. Researchers conducted two series of workshops (W#1 and W#2) (Fig. 1). Each workshop was conducted with two different groups (A and B). Group A was comprised of experts (regional actors and researchers), while Group B involved local actors and stakeholders (along with the researchers). The local actors, and experts or regional actors are all involved in the wine sector (a production cooperative, the union of Protected Designation of Origins (PDO), the advisory and extension services, and public administrators in charge of water, subsidies, or agriculture - Table 1). The hypothesis behind the decision to work with these two groups separately was that they would provide different but complementary perspectives. We hypothesized that experts would propose more innovative levers and strategies that would be less dependent on context, while stakeholders and local actors would be more likely to suggest ideas that evoke the real-world constraints and obstacles of "doing business," thus helping to highlight their technical and organizational feasibility and acceptability.

### 2.2. Study area

The Rieutort watershed (France, NUTS-3 Hérault) was chosen for the case study (Fig. 2). The watershed, covering 45 $km^2$, is dominated by vineyards (1200 ha) and presents a diversity of landscapes (e.g., differences in slope, grape varieties, planting density, soils). The climate is Mediterranean, with an overall rainfall gradient increasing from the South to the North (2000–2019 mean annual rainfall of 581 mm and 705 mm, respectively - weather stations of Puisserguier and Cabrerolles

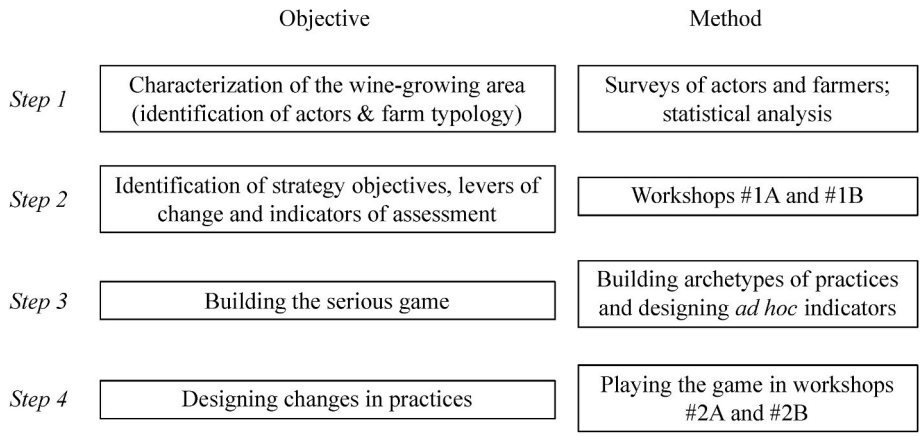



**Fig. 1.** Overview of the overall approach in four main steps.

**Table 1**
Participants in the different study steps. For step 1, only winegrowers with complete survey information were included.

| | | Step 1 | Step 2 | | Step 4 | |
|---|---|---|---|---|---|---|
| Group | Participant category: representatives of | Interview | W#1A | W#1B | W#2A | W#2B |
| Experts | Public Administration in charge of water resources management | – | 1 | – | 1 | – |
| | Vine Technical Institute | – | 1 | – | 1 | – |
| | Union of PDO | – | 2 | – | 1 | – |
| | Extension and advisory services | – | 1 | – | 1 | – |
| | Union of defence against harmul organisms | – | 1 | – | 1 | – |
| | Regional public administration in charge of subsidies | – | 1 | – | – | – |
| | Regional public administration in charge of agriculture | – | 1 | – | 2 | – |
| Local actors | Farmers FT1 | 11 | – | 6[a] | – | 4[a] |
| | Farmers FT2A | 4 | – | 1 | – | – |
| | Farmers FT2B | 4 | – | 3 | – | 2 |
| | Farmers FT3 | 3 | – | – | – | – |
| | Local cooperative winery | 2 | – | 1[a] | – | 1[a] |
| | Extension and advisory services | 0 | – | 1 | – | – |
| | Local seller of phytosanitary products | 1 | – | – | – | – |
| | Union of Saint Chinian PDO | 1 | – | 1 | – | 1 |
| | Union of Faugères PDO | 1 | – | – | – | – |
| | Local public administration in charge of water resources management | 1 | – | 0 | – | 2 |
| Both groups | Researchers | 0 | 6 | 2 | 6 | 3 |
| Total | | 28 | 14 | 14 | 13 | 12 |

[a] Note that the representative of the local cooperative winery was also a winegrower (FT1). FT: Farm type (see details in §3.1), PDO: Protected Designation of Origin.

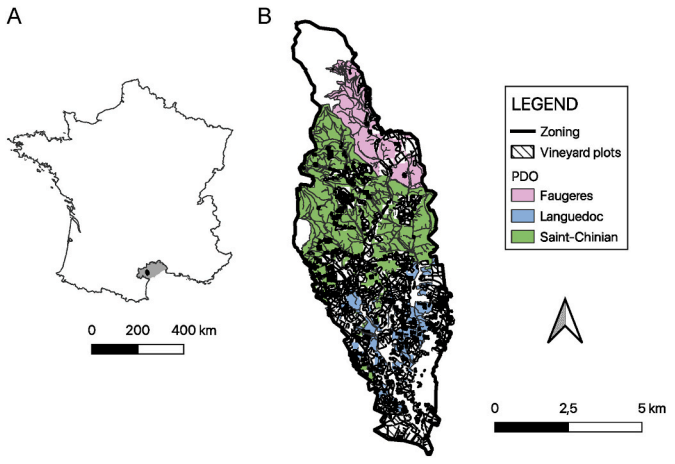



**Fig. 2.** Case study watershed. A. Location of the Rieutort watershed in Metropolitan France (in grey the Hérault NUT3 region; in black the watershed); B. Rieutort watershed, with vineyard fields (hatched) and associated Protected Designation of Origin (PDO), white fields in the South correspond to Protected Geographical Indication (PGI). Sources: EPTB Orb et Libron, 2020

operated by Météo-France). The vineyard area is divided into four main classifications of grape production: to the south there is one Protected Geographic Indication (PGI) called Pays d'Oc, and to the north there are three PDOs called Languedoc, Faugères, and Saint-Chinian (Fig. 2). Faugères and Saint-Chinian are part of the Languedoc PDO. These classifications of production are associated with different specifications in terms of the cultivars accepted, planting density, and maximum yield, which in practice leads to a gradient of yield performance, roughly decreasing from the South to the North. In 2020, there were 136 winegrowers with at least one field in the watershed (data EPTB Orb et Libron, 2020).

### 2.3. Characterization of the current situation

The characterization of the current situation was performed with individual surveys of winegrowers and local actors (Table 1). It included three categories of information: (1) the general characteristics of the watershed; (2) the description of current farms; and (3) the different institutions, structures, and people involved in the agricultural development of the Rieutort watershed. Most of the local actors were known from previous research work (Naulleau et al., 2022); contacts for most winegrowers were provided by the local actors. Additional winegrowers were identified through snowballing.

In-depth interviews were conducted with 26 winegrowers, but

complete information was assembled for only 22 of them (representing 33% of the watershed vineyard area). These interview surveys focused on specific information: (1) farm structure (e.g., area, organic or other environmental labels, cooperative or private wine production); (2) cropping practices (e.g., cultivars, soil treatments, pesticide applications and other means of pest and disease control), (3) workforce (permanent and temporary), and (4) materials (for pesticide application and tillage operation).

The information collected was used to build a farm typology with a Multiple Factor Analysis (MFC), followed by an Ascendant Hierarchical Classification with the R packages FactoMineR (Le et al., 2008)) and factoextra (Kassambara and Mundt, 2020). All surveyed variables were used in MFC, except those on pesticide applications, which were used as supplementary variables. One of the three farm types identified by the MFC was split into two types because the diversity of pesticide treatment plans was too large to characterize the practices of one single winegrower.

### 2.4. Co-definition of practices and evaluation criteria

This step was performed in the first series of workshops, W#1A with experts, and W#1B with local actors and stakeholders (4 h for each workshop). W#1A was conducted remotely (a Covid security measure), using Padlet© online tool. Two boards were prepared and sent to participants beforehand: (1) one for listing the objectives for reducing the impacts of pesticides, together with the criteria to assess the achievement of those objectives, and (2) one for listing the levers of action. A structural proposition was made for each board: (1) according to the targeted element (water, air, soil) or to the relevant lever of change for the objectives and criteria, and (2) according to the scale of lever application (field, farm, agroecological infrastructures, landscape mosaics). Workshop participants could add a new category, either before or during the remote workshop. After a presentation of the workshop structure, all the participants worked together to fill the first board with objectives and criteria; to fill the second board, the participants were split into two groups in order to gather as many levers of change as possible.

W#1B was structured in three parts: (1) presenting the results of the farm typology and discussing them with the participants; (2) listing the levers of changes; (3) comparing and discussing the levers of changes identified by experts in W#1A, but not proposed here. Each participant proposed his levers on separate post-its (three to five per participant). Levers were then grouped and collectively discussed.

### 2.5. The CAPIPP game

The game consists of seven elements (Fig. 3): (1) the description of typical farms, (2) the description of the potential levers of change for each farm, (3) a change choice sheet for each farm type and for each of the three blocks (groups of plots) of the farm, (4) a simplified representation of the watershed, (5) information panels on processes and levers of action, (6) additional information, and (7) a calculator to assess *ex ante* the potential impacts of the practices at different scales.

The game, which is played in two rounds, includes two types of roles: winegrowers and a public manager. Each winegrower is in charge of a typical farm. The public manager evaluates the winegrowers' requests for subsidies (based on a watershed budget of 300,000€ for 1200 ha of vineyard and current subsidies communicated by EPTB) and the evolution of the potential impacts. Those evaluations are noted at the end of the first round, and in the second round the public manager can suggest new changes in practices and the possibly of new subsidies.

There are either five or six farms in the game, each belonging to one of the four farm types derived from the initial survey. More farms of Farm Type 1 were included in the game because this type reflected a

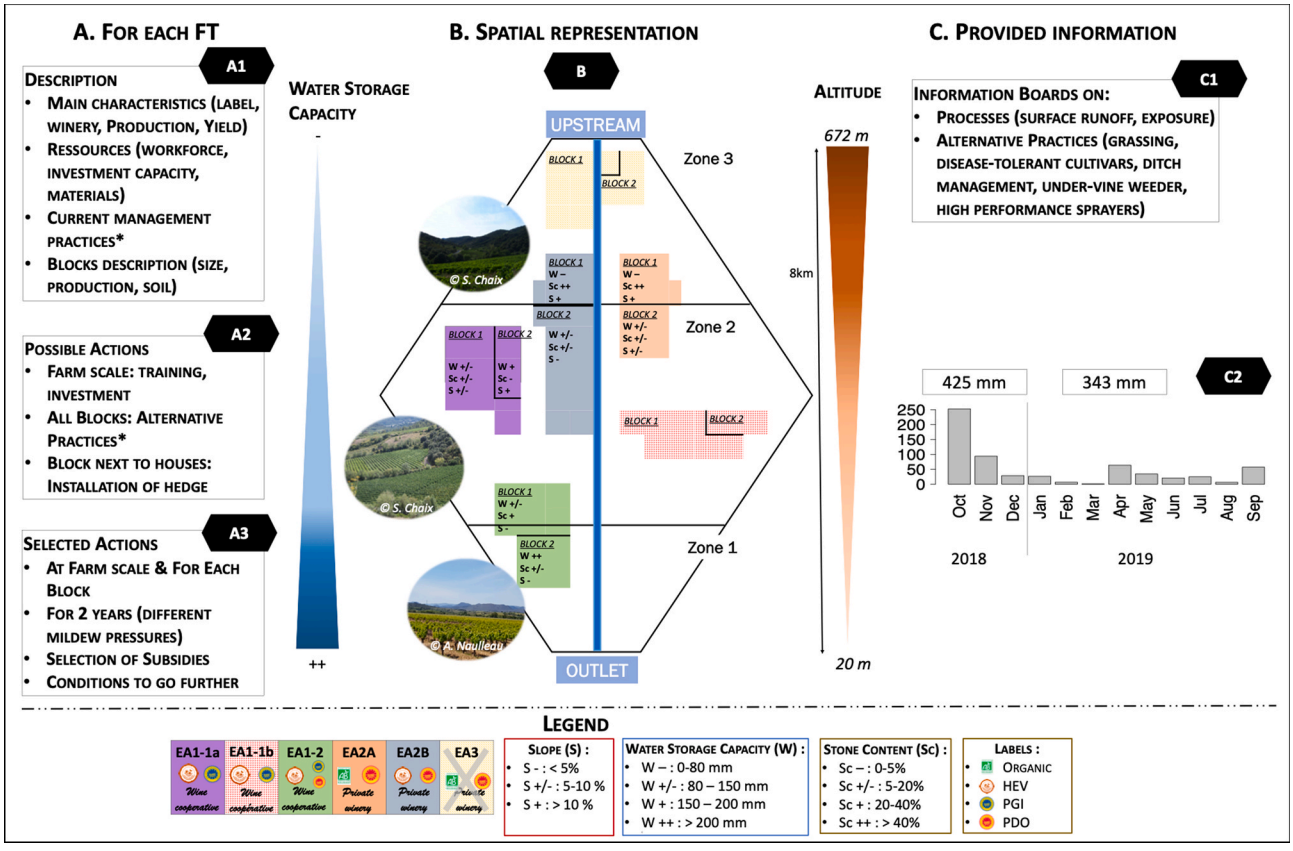



**Fig. 3.** Overview of the different components of CAPIPP game. FT: Farm Type; * See Table 4 for current and alternative practices.

larger portion of vineyard area (see §3.1). Farm type 3 was not in the game because it is associated with a niche market representing minimal environmental impacts. Each included farm was composed of three blocks, two of the blocks could differ in terms of size, soil, slope, or production (PGI, PDO); the third block (1 ha) was characterized as being next to houses. The farm's description (structure, resources, current management practices) was used to describe the role to players. The description was based on the dominant characteristics of each farm type, namely structure and resources and most management practices (tillage, herbicide, fungicide, and insecticide applications that presented a low within-group variability). To build the current management practices for a given farm type, we calculated the evaluation indicators (detailed below) for all farms of the same type and then chose the farm exhibiting values within the third quartile of the indicator distribution. When no farm within a farm type presented this profile (e.g., for Farm Type 1), we chose the one presenting values above the median for most indicators to be the representative farm.

The levers of change differed between farm type, depending on their initial state and characteristics. Theses levers were taken from the lists made by experts and local stakeholders in W#1A and W#1B. For levers like fungi management, we included control strategies developed in local field experiments (Reasoned +, Organic Agriculture +; see Appendix A for details). Each player in the winegrower role had to choose which changes to make at two scales: (1) at the farm scale, training about the use of levers, different sprayers, and ditch management; (2) at the block scale, row management, inter-row management (soil tillage and grassing), fungi and insect management, headland management, and type of spraying, for each block (plus installation of hedgerow for the block next to houses). Choices also had to be made for years subject to different mildew pressure (low or high for years similar to 2019 or 2020, respectively). Some changes were eligible for subsidies involving a 5-year commitment to a change of practice (e.g., grassing of inter-rows); others could not be chosen without asking for specific training. Note that we considered a unique cultivar, Grenache, currently the most common grape variety in the area (27% of vines).

The simplified representation of the watershed used different sets of information: the farm typology and spatial location of farm types, a 1:100,000 soil map (Bonfils, 1988) refined by local soil surveys, and a slope map derived from a 5 m resolution digital elevation model. Because the farms covered by the survey represented only one third of the total vineyard area, we mobilized a local actor familiar with the watershed, to allocate each of the non-surveyed farms to one of the farm types. This allowed us to cover 75% of the vineyard area. Based on other information at farm scale (e.g., private/cooperative winery, type of production, size; source EPTB Orb et Libron, 2020), we could allocate most of the remaining non-surveyed farms, ultimately achieving 95% coverage.

We provided information panels on processes (e.g., surface runoff influenced by soil treatment) and levers of action (e.g., disease-resistant cultivars, grassing, ditch management - Fig. 3) to help workshop participants with their decision-making. Since changes in practices depend on other factors, we also provided additional information: (1) rainfall conditions for the two crop years (2018–2019 and 2019–2020); (2) available subsidies, conditions, and value per hectare; and (3) a listing of products and doses applied for each possible plant protection practice (current and alternatives).

At the end of each round, the calculator uses a set of indicators to assess the potential impacts of new practices as compared to those of the initial practices. This assessment is made at both farm type and watershed scales. Upscaling from farm types to watershed is based on extrapolation coefficients using the 95% cover of the vineyard area. The choice of indicators is based on the list of criteria developed by experts in W#1A.

At both scales, there are two types of evaluation indicators: (1) indicators of use (TFI- Treatment Frequency Index, TFI excluding biocontrol products, and TFI with Carcogenic, Mutagen, and Reprotoxic products); and (2) indicators ranking pesticide treatment plans according to their potential impacts on neighboring residents, earthworms, pollinators, aquatic organisms, operators, and water quality. The indicators of potential impacts were kept simple to allow their immediate assessment and visualization during the game session. The assessment was made in an application built with the RShiny package (Chang et al., 2021).

For earthworms, pollinators, aquatic organisms and operators, the impact indicators were the sum of the estimated toxicity contribution of all applied substances as measured by the quotient of their applied amount to their specific ecotoxicological endpoints. For residents, the impact indicator also considered the efficiency of the spraying technique and the volatility of the applied substances. For drinking water catchment, the indicator was aimed at ranking the risk of contamination of water resources by summing the cross-products between the mass of each substance and its Groundwater Ubiquity Score (GUS) as proposed by Gustafson (1989). All formulae and data sources are provided in Appendix B.

At watershed scale, the evaluation indicators also included other indicators of use that the experts considered relevant at this scale, such as the number of anti-drift sprayers, prevalence of Organic Agriculture, absence of herbicide use on rows, and prevalence of disease-resistant cultivars.

### 2.6. Playing the game

CAPIPP was played in two different sessions (W#2A and W#2B). The organization of the workshop was the same for the two sessions, except the farm typology was only presented to the groups of experts. All components of the game were presented to all the participants (Fig. 3). Participants were then assigned a role: either a public manager or a winegrower of a specific Farm Type. After the 1st round (choice of practice changes and assessment), the public manager gave feedback on the possibility of granting requested subsidies and on the expected impact reduction from proposed practice changes. Based on these comments and suggestions (e.g., additional subsidies for changing spraying devices, increasing tillage to replace chemical weeding), winegrower players could decide whether or not to further modify their practices and obtain new assessments (round 2). Finally, a debriefing was organized around three themes to bring real-world issues to the fore and collect additional information about decisions: (1) chosen practice changes, their realism, their feasibility without subsidies, and their validity for other cultivars; (2) alternatives not chosen by the participants (purchase of fields by public authorities, and practices such as grassing or organic that were chosen by the animator after the second round); and (3) the assessment of the game's key points and positive or negative aspects. In W#2B, participants were asked to vote on a consensus scale for the chosen wine-growing practices and the field purchase by public authorities. A discussion allowed participants to express their views about the other points. In W#2A the debriefing was an open discussion.

We adopted different strategies to assign the roles, but when possible, we tried to assign a role similar to the participant's real-life experience. For example, in W#2A, the role of public manager was assumed by the person from the public administration in charge of water management, paired with a researcher specialized in water resources. Because all participants in W#2A did not have practical field management experience, we established pairs of participants to balance the distribution of organizations, knowledge, and skills. In one exception, the participant was both a researcher and a winegrower in the real life.

We did not form pairs in W#2B because the real winegrowers make their decision alone in the "real life". We did, however, divide W#2B into two groups after the game presentation, leading to two parallel sessions (W#2B1 and #W2B2). The role of public manager in W#2B1 was assumed by two members of the local administration in charge of water (they were not comfortable assuming this role alone). In W#2B2,

the role was assumed by the same researcher as in W#2A. In the role of winegrower, the farm type assigned to each participating winegrower matched the type of their actual farm. To maintain this convention, one farm type (FT2A, organic) was not played in W#2B because the organic winegrowers were not able to attend the workshops. In addition, because of the larger area of farm type 1-1 (FT1-1), it was divided and played by two winegrowers (FT1-1a and FT1-1b) in W#2B1. In W#2B2, FT1-1 was played by a PDO technical adviser.

## 3. Results

### 3.1. Characterization of current situation

The final typology, based on the survey data from 22 winegrowers, led to four distinct farm types (Table 2). Only Farm Type 1 (FT1) produced mostly PGI grapes in bulk wine production with the High Environmental Value (HEV label). It also presented the highest yields, and the most extensive use of pesticides (Table 2). Each of the three other Farm Types (FT2A, FT2B, and FT3) involved fewer winegrowers and produced PDO grapes. FT2A and FT3 used less pesticides, and both productions have an Organic Agriculture label (OA).

Classifying all farms according to this typology led to 40%, 25%, 7%, 17 and 11% of the vineyard area belonging to FT1-1, FT1-2, FT2A, FT2B and FT3 farm types, respectively.

### 3.2. Indicators and levers of changes

#### 3.2.1. Objectives and indicators

The experts in W#1A cited two types of objectives to reduce the impacts of pesticides: (1) means-oriented, and (2) results-oriented. The means-oriented objectives were related to (1a) "external factors" (i.e., they depend on actions at a higher scale, such as subsidies or changes in PDO rules), and (1b) internal farm management (Table 3). For external factors, the objectives were to orient subsidies towards the use of anti-drift sprayers, to increase dissemination actions, and to increase the cultivation of disease-resistant cultivars by including them in PDO specifications. For internal farm management, the objectives were to change ditch management, increase inter-row grassing, and use less fungicides. Results-oriented objectives involved (2a) the environmental quality of water, air, and soils, and (2b) farming performance. The environmental quality objectives were to decrease the level of pesticides in water resources, limit pesticide exposure among winegrowers and local residents, and limit ecotoxicological impacts. The farm scale objectives involved production and economic performance, and winegrower working conditions. All of the objectives were associated with indicators to evaluate their attainment. The indicators were aimed at assessing means (e.g., area with no herbicide), or impact (e.g., not meeting standards for water quality, risks for residents, yield loss) (Table 3). Some of the indicators had a dynamic aspect such as "decreasing with time" or "cumulative with respect to time". Those could not be considered in the CAPIPP calculator, which is designed to remain simple and can only calculate static indicators. Similarly, indicators related to farm economics, or erosion were not considered in the game (Table 3).

#### 3.2.2. Levers

While formulating the objectives, the experts also mentioned several levers of action to reduce the impacts of phytosanitary products. These levers were elaborated in W#1B by winegrowers and local stakeholders (Appendix C). The suggested levers for action and alternative practices related to different spatial scales ranging from the plant (technical levers) to large scale European policies (subsidies). The levers also referred to different domains. Some were pest-specific (e.g., disease-resistant cultivars to manage fungi, Group of Defense against Harmful Organisms (GDHO) for controlling insects and reducing treatments, mechanical weeding). Others were related to economics (e.g., subsidies, labels), to the advisory system (e.g., training), to machinery (e.g., anti-drift sprayers, shared materials), to infrastructures (washing areas), or to production orientation and landscape (e.g., diversifying production, ditch management, development of organic agriculture). Many levers were proposed by both groups (W#1A and W#1B), although levers at the landscape scale came mainly from the group of experts. Most of these levers could be included as alternatives in the game (Appendix C).

Some of these alternatives differed between Farm Types in terms of row, inter-row, and fungi management (Table 4) because they had different initial practices (e.g., FT2A has an Organic Agriculture label). Farm Type 1 had the most alternatives choices for fungicide management because it initially applied the most fungicides. But some alternatives were common to all farm types: hedgerow and headland management, installation of a ditch, and number of insecticide treatments (organic management used different products than those used by other types of management) (Table 4).

### 3.3. Individual strategies and their estimated impacts at farm scale

In the game sessions, training on sprayer adjustment was requested by all winegrower players in W#2A, and by 81% of all players in W#2B. The second training most frequently requested by winegrower players (45%) concerned the management of fungal disease, with a slightly different focus: for optimization, for using biocontrol products, and for

**Table 2**
Dominant characteristics of the four farm types and associated representation in the game.

| CURRENT SITUATION | Farm types | FT1 | | FT2A | FT2B | FT3 |
|---|---|---|---|---|---|---|
| | Nb. surveyed | 11 | | 4 | 4 | 3 |
| | Nb ha/worker[a] | 10 | | 7 | 8 | 7 |
| | Harvest type | Mechanical | | Mixed | Mechanical | Manual |
| | Sale type | Bulk | | Bottles | Mixed | Bottles |
| | Label | HEV | | OA | Mixed | OA |
| | Designation[b] | PGI | | PDO | PDO | PDO |
| | Mean yield (hl/ha) | 72 (64)[c] | | 36 | 39 | 18 |
| | TFI without biocontrol products for 2019; 2020 | 9.9; 12.8 | | 3.4; 5 | 8; 10.8 | 2.6; 5.3 |
| | TFI total for 2019; 2020 | 10.9; 14.5 | | 6.4; 7.4 | 10.3; 13.9 | 7.1; 11.4 |
| | Median size (ha) | 21 | | 19 | 30.5 | 16.5 |
| CAPIPP game | Farm types[d] | FT1-1 | FT1-2 | FT2A | FT2B | FT3 |
| | Size (ha) | 21 | 21 | 18 | 31 | 21 |
| | Extrapolated area (ha)[5] | 446 | 286 | 74 | 188 | 130 |

[a] Excluding harvest.
[b] Dominant designation (>75% of farm area).
[c] In parenthesis is the mean yield when PDO vineyards are included.
[d] FT1 was divided in two (FT1-1 and FT1-2) because of the large area involved. FT: Farm type; HEV: High Environmental Value; OA: Organic Agriculture; PGI: Protected Geographic Indication; PDO: Protected Designation of Origin.

**Table 3**
Co-defined objectives and indicators of evaluation as defined by experts. The objectives and indicators used in the CAPIPP game are indicated in italics.

| Type of objective | Objective | Indicator | Scale |
|---|---|---|---|
| Depending on external factors | Purchase of anti-drift sprayers and sprayer adjustments | Number and area covered with anti-drift sprayers; % area with optimized sprayer; mean drift rate | Watershed |
| | Restructuring of grape varieties with disease-resistant ones | Evolution of areas (specific focus on vulnerable areas, i.e., next to houses and waterways) | Watershed |
| | Dissemination, training, and support | Number of winegrowers in reasoning groups, or subscribers to phytosanitary bulletins | Watershed |
| Internal to farm | Ditch management | % of ditch areas with specific management (e.g., burning); Pesticide retention rate | Watershed |
| | Grassing of inter-rows | % of grassed vines; % of wine with no herbicide on inter-row; % of wine with no herbicide | Watershed |
| | Decrease of fungicides | Total fungicide TFI, and without biocontrol products; Ecotoxicity | Field; farm |
| Regarding water quality | Reduction of the levels of phytosanitary products in drinking water catchments | Maximum level < norm; Decreasing with time | Watershed |
| | Reduction of pesticides dispersal in surface water | Maximum level < norm; Decreasing with time | Watershed |
| | Washing areas | % of winegrowers using it | Watershed |
| Regarding air quality | Limit the exposure of local residents | % of area with concentration < threshold (with focus on areas next to houses); Number of days with concentration > threshold; Level of satisfaction of residents | Watershed |
| | Exposure of the winegrower and the operator | % use of individual protective equipment; Reduction in cumulative exposure | Field; watershed |
| Regarding soil quality | Limit soil storage and ecotoxicological impact | % area with level < threshold; Toxicity; Microbial life | Field; watershed |
| | Preserve soil quality | Erosion and soil compaction | Field |
| Regarding farm performances | Limit the loss in income | % of loss | Farm |
| | Limit yield loss | Water competition due to grassing | Field; farm |
| | Winegrower working conditions | Working time; Energy consumption | Farm |

**Table 4**
Initial and alternative practices for the different farm types (FT).

| Farm type | FT1-1; FT1-2 and FT2B | | | FT2A | | | All Farm types | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Operation | Row | Inter-row | Fungi | Row | Inter-row | Fungi | Insect[a] | Hedgerow installation[b] | Headland | Ditch |
| **Initial practices** | 2 glyphosate +1 anti-germination | 4 plowing (bare soil) | Preventive[c] | 2 under-vine weeder operations | 2 plowing | OA | 2 treatments | No | Chemical | No |
| **Alternative practices** | 1 glyphosate; 0 glyphosate[d] | 3 plowing; 2 plowing[e]; 1 plowing + 1 glyphosate; 1/4 grassed; 1/2 grassed; all grassed[f] | 0 CMR; AO[g]; Reasoned +[h]; AO+[8]; Disease-resistant cultivars[g] | – | 1/4 grassed; 1/2 grassed; all grassed[f] | AO+[8]; Disease-resistant cultivars[g] | 1 treatment | Yes | Grassed | Mowing; Clearing; Burning; Chemical |
| **Condition** | Buy an under-vine weeder | – | Training for Reasoned+ and AO+; Planting for disease-resistant cultivars | – | – | Training for AO+; Planting for disease-resistant cultivars | Join a GDHO | – | – | – |
| **Possible subsidy** | 50% of the weeder price; AEM of 0 glyphosate | AEM for grassing | AEM for OA, Reasoned +, and AO+ | – | AEM for grassing | AEM for AO+ | Subsidy for GDHO creation | AEM | – | – |

OA: Organic Agriculture; CMR: Carcinogenic, mutagen and reprotoxic active substances; AEM: Agro-Environmental Measures (subsidies from the Agricultural Common Policy); GDHO: Group of Defense against Harmful Organisms.

[a] Chemical products used for insect management differ between conventional and organic.

[b] Only for block next to houses.

[c] Note that the treatments are identical for FT1-1 and FT1-2, but differ for FT2B.

[d] Both alternatives involve mechanical weeding.

[e] Decreasing the amount of plowing involved abandoning autumn plowing.

[f] In alternatives with grassing, plowing continues on the non-grassed rows, and mowing on the grassed ones.

[g] OA alternative corresponds to the initial practices of FT2A.

[h] These alternatives are based on on-station experiments (Appendix A).

Organic Agriculture or Reasoned+. The last training requested in more than one session concerned the use of the under-vine weeder. All these trainings were requested by the group of experts in W#2A, but additional trainings were requested in W#2B: training for alternatives to pesticides, composition and knowledge on pesticides, weeding and plowing, and insect observations and GDHO. Comparing the results of the three sessions did not highlight specificities per Farm Type.

At farm scale, only players of FT1 in W#2B decided to invest in equipment without subsidies. The winegrower playing FT1-2 in W#2B2 wanted to invest in a roadside crusher "to manage the borders and the grassed rows." The winegrowers playing FT1-1a and FT1-1b in W#2B1 wanted to invest in anti-drift weeding ramps. Materials subject to

possible subsidies were requested by winegrowers in W#2A and W#2B1. In the second round, all winegrower players in W#2A (experts) except FT1-1 asked for subsidies to change their sprayer (air-blast sprayer for FT1-2, confining and recycling sprayer for FT2A and FT2B). In W#2B1, two winegrower players (FT1-1a and FT1-2) asked for a subsidy to buy an under-vine weeder, and one player (FT1-2) asked for a subsidy to buy a confining and recycling sprayer. No subsidy for material was requested in W#2B2.

Requests for subsidies at field scale differed largely between the groups of players, with more ambitious changes proposed by the group of experts (W#2A). In this group, two winegrower players (FT1-1 and FT2B) asked for subsidies to stop using glyphosate on all their vineyards; FT1-1 even decided to convert to Organic Agriculture. FT2A and FT2B asked for subsidies to grass the inter-rows in part of their vineyards (1 in 2, or 1 in 4 inter-rows). Among the groups of winegrowers and local stakeholders, only players of FT2B of W#2B1 and FT1-1 in W#2B2 asked for subsidies to grass the inter-row (1 in 4 on some blocks). All winegrower players in W#2B1 asked for a subsidy to install a hedgerow on the border of their block located next to houses. All winegrower players in the local group decided to install (or maintain) hedgerows for protection against drift, but only two farm-type players in the expert group made this choice. Out of the 11 winegrower players from the three game sessions, seven decided to mow their ditches, one to burn them, and three decided not to manage them.

#### 3.3.1. 2019 type year (low mildew pressure)

Among the alternative practices chosen, all winegrower players decided to optimize the settings of their sprayer (except FT1-2 in W#2B2). In the three sessions, all winegrower players decided to grass the headlands in all their farm areas, and most of them decided to be part of a GDHO, making it possible to reduce the number of insecticide treatments. All winegrower players decided to stop autumn plowing, which can lead to erosion during heavy rainfall. The choices concerning changes in inter-row grassing and herbicide and fungicide treatments were more diverse.

For farm type results, we detail Farm type 1-1, and then give the main elements for the other farm types (see details in Appendix D). For Farm type 1-1, three of the four winegrower players chose to use less herbicides for row management on the block next to houses (0 or 1 glyphosate treatment, only the expert player chose 0). Two of the winegrowers playing FT1-1 also chose to use less herbicides for their two main blocks. In terms of inter-row management, only one player decided to grass in all three blocks. For fungi management, two players from the local group chose the same management for all three blocks (0 CMR, or Reasoned+), while the two other players opted for more environmentally friendly management alternatives on the block next to houses (Table 5). A more environmentally friendly practice on one pest (e.g., weed management) did not necessarily correlate to more environmental-friendly practices for other pests. Practice changes led to different impacts for the block next to houses, and at farm scale, for the different players, except for the indicator on CMR (0 for all). All indicator values decreased for the block next to houses, but not at farm scale (Fig. 4). The profile of lowest impact for FT1-1 was obtained for the expert group (W#2A) for the block next to houses (due to disease-resistant cultivars). At farm scale, the profiles of indicators were similar among the local players of W#2B1 for all blocks (similar choices). The profiles of indicators for W#2A (experts) and W#2B2 differed mainly because of changes in fungi management (Table 5). The potential impact on pollinators was higher for the expert player (W#2A) choosing organic agriculture for blocks 1 and 2, due to the application of sulfur to treat powdery mildew (as this indicator considers the quantity applied, Appendix B).

For Farm Type 1–2, one local player (W#2B2) chose the same alternative for all three blocks (1 herbicide on the rows, 3 plowing on the inter-rows, and disease-resistant cultivars) (Appendix D-1). For all three blocks, the expert player (W#2A) chose grassing for 1 in 2 inter-rows, but no change in row management. He opted for more environmentally friendly management of fungi on the block next to houses. The last player (W#2B1) did not change fungi management and only reduced herbicide use and plowing (inter-row management) for the block next to houses.

Farm Type 2A was played in the group of experts only. This pair of players changed only the inter-row management, choosing to grass 1 inter-row in 4 for the block next to houses and for block 2 (Appendix D-2).

For Farm Type 2B, two of the three players opted for more environmentally friendly alternatives for the block next to houses, although on different practices. The pairs of experts (W#2A) chose to manage fungi differently. They stopped glyphosate in the rows and chose to grass 1 in 2 inter-rows for all blocks. The local player in W#2B1 differentiated row and inter-row management and applied the 0 CMR alternative on all blocks. The local player of W#2B2 opted for the same management on all three blocks: 1 glyphosate on rows, 3 plowing inter-rows, and Reasoned + for fungi management (Appendix D-3).

Overall, both groups of players in workshops #2A and #2B (experts and local actors) made more ambitious changes for the block next to houses than for the two other blocks (Table 6). For all three blocks, changes in grassing on the inter-rows, row management, and fungi

**Table 5**
Farm Type 1-1 changes in practices for a 2019 type year (low mildew pressure) for the three sessions of the CAPIPP game's 1st round.

| | Session | Experts' session | Winegrowers' session 1 | | Winegrowers' session 2 |
|---|---|---|---|---|---|
| Block | Practice | W#2A | W#2B1-1 (FT1-1a) | W#2B1-2 (FT1-1b) | W#2B2 |
| Next to houses | Hedge | No | Yes | Yes | Yes |
| | Row management | 0 herbicide | 1 glyphosate | 1 glyphosate | Initial |
| | Inter-row management | 3 plowing; 1/2 grassed | 3 plowing; 1/4 grassed | 3 plowing; 1/2 grassed | 2 plowing |
| | Fungi management | Disease-resistant cultivars | Reasoned + | 0 CMR | OA+ |
| | Insect management | GDHO | GDHO | GDHO | GDHO |
| | Headland | Grassed | Grassed | Grassed | Grassed |
| | Sprayers | Optimized | Optimized | Optimized | Optimized |
| 1 | Row management | 0 herbicide | Initial | 1 glyphosate | Initial |
| | Inter-row management | 3 plowing | 3 plowing | 3 plowing; 1/2 grassed | 3 plowing; 1/4 grassed |
| | Fungi management | OA | Reasoned + | 0 CMR | Reasoned + |
| | Insect management | GDHO | GDHO | GDHO | GDHO |
| | Headland | Grassed | Grassed | Grassed | Grassed |
| | Sprayers | Optimized | Optimized | Optimized | Optimized |
| 2 | Row management | 0 herbicide | Initial | 1 glyphosate | Initial |
| | Inter-row management | 3 plowing; 1/2 grassed | 3 plowing | 3 plowing; 1/2 grassed | 2 plowing |
| | Fungi management | OA | Reasoned + | 0 CMR | Reasoned + |
| | Insect management | GDHO | GDHO | GDHO | GDHO |
| | Headland | Grassed | Grassed | Grassed | Grassed |
| | Sprayers | Optimized | Optimized | Optimized | Optimized |

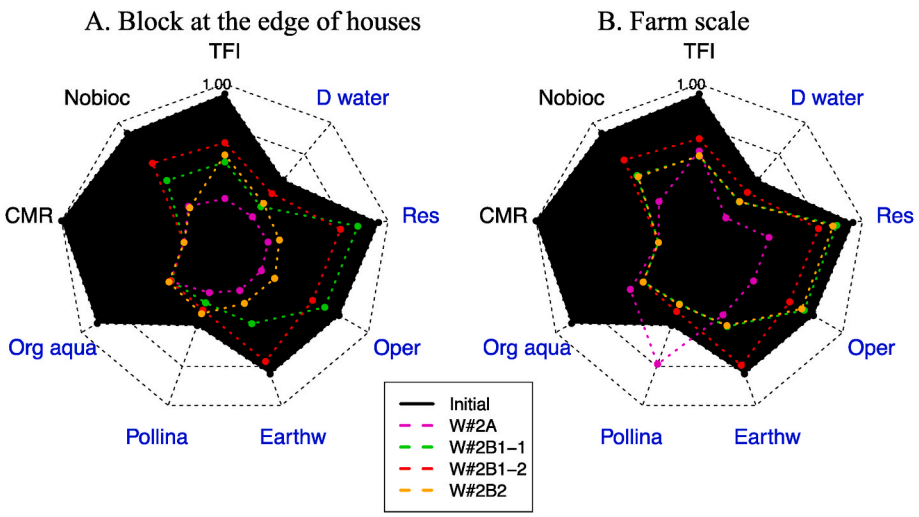



**Fig. 4.** Impacts of Farm1-1 changes in practices for a 2019 type year (i.e., low mildew pressure) for the three sessions of the CAPIPP game's 1st round. A. Results for the block next to houses; B. Results at farm scale. Labels of means-based indicators are in black text (TFI – Treatment Frequency Index; Nobioc – TFI without biocontrol products; CMR – Carcogenic Mutagen and Reprotoxic products), and labels of impacts-based indicators are in blue text (Org aqua – Aquatic organisms; Pollina – Pollinators; Earthw – Earthworms; Oper – Operators; Res – Residents; D water – Drinking water catchment). Note that workshop W#2B was split into two groups of players (W#2B1 and W#2B2), and FT1-1 was played twice (FT1-1a and FT1-1b, Table 5) in W#2B1 (named in figure W#2B1-1, and W#2B1-2). For each indicator, minimum and maximum spider values correspond to the minimum and maximum value when considering all combinations of current and alternative practices.

**Table 6**
Synthesis of the practices (as a percentage of the number of players) in 2019 type year chosen by players of W#2A and W#2B for the 1st round of the CAPIPP game. Note that Farm Type 2A (the only type with an existing organic label) was not considered here, because it was not included in the game in session W#2B.

| | | All blocks | Blocks 1 and 2 | | Block next to houses | |
|---|---|---|---|---|---|---|
| Management | Option | Initial | W#2A | W#2B | W#2A | W#2B |
| Fungi options | Conventional[a] | 100 | 0 | 14.2 | 0 | 14.2 |
| | 0 CMR | 0 | 33.3 | 28.6 | 0 | 28.6 |
| | Reasoned+ | 0 | 33.3 | 42.9 | 0 | 28.6 |
| | OA | 0 | 33.3 | 0 | 0 | 0 |
| | OA+ | 0 | 0 | 0 | 33.3 | 14.3 |
| | Disease-resistant cultivars | 0 | 0 | 14.3 | 66.7 | 14.3 |
| Options for grassing on inter-row | No | 100 | 16.7 | 78.6 | 0 | 57.1 |
| | 1/4 | 0 | 0 | 14.3 | 33.3 | 28.6 |
| | 1/2 | 0 | 83.3 | 7.1 | 66.7 | 14.3 |
| Options or row | Conventional | 100 | 33.3 | 57.1 | 33.3 | 14.3 |
| | 1 herbicide | 0 | 0 | 42.9 | 0 | 85.7 |
| | 0 herbicide | 0 | 66.7 | 0 | 66.7 | 0 |
| Total number of blocks | | 9 | 6 | 14 | 3 | 7 |

[a] Note that fungi management is the same for FT1-1 and FT1-2 but differs for FT2B.

management were also more ambitious in the group of experts than in the local groups. The experts chose more organic management, more grassing (1 in 2 inter-rows), and no herbicide use (Table 6).

#### 3.3.2. 2020 type year (high mildew pressure)

Few adjustments to selected alternative practices were made by players for the case of higher mildew pressure. No adjustment was made in the expert group. Adjustments were made by local players in W#2B1: grassing inter-rows in some blocks (FT1-1 and FT1-2), decreasing plowing (FT1-1, FT1-2, and FT2B), and removing the option of environmentally friendly fungi management (from Reasoned + to initial practices). One local player (FT2B) of W#2B2 also modified the choices for fungi management, selecting 0 CMR in the case of high pressure (instead of Reasoned+).

### 3.4. Aggregated impacts at the watershed scale

We detail here only the most important levers chosen, and not chosen, when scaling up to the watershed. At the watershed scale, no anti-drift sprayer was bought for use in the first round of the game. Disease-resistant cultivars were cropped on a large part of the area in W#2B2 only (due to the large area exploited by FT1-2 farms which chose this fungi management option). The group of experts strongly opted for 0 pesticide use on rows, contrary to the groups of local actors (Table 7).

Changes at farm scale led to a range of new indicator values, either higher or lower than the initial ones (Appendix D). However, all sessions involved a decrease in almost all indicators for both years at the watershed scale (Fig. 5). The only exception was for the indicator on pollinators, which increased in 2019 due to the intensive use of sulfur in Organic management for fungi chosen by the FT1-1 player (Table 5, Fig. 4). The game sessions leading to the largest decrease in potential impacts of pesticide use were those with the group of experts (W#2A, except for pollinators and aquatic organisms), and with local actors (W#2B2). This was largely due to the choice of FT1-2 to change its fungi management in favor of disease-resistant cultivars. In all sessions but W#2B1, practice changes led to 0 CMR active substances over the watershed. The ranking of indicator values between sessions differed for some indicators, the ranking of the water indicator (taking into account mobility properties of active substance) was opposite to the one of Aquatic organisms indicator (Fig. 5). Ranking of means-based indicators did not differ from those of results-based indicators.

### 3.5. Discussion with the public manager players

In all sessions, public manager players began by acknowledging the efforts made by winegrower players and the resulting decrease in potential environmental impacts. Along with the group of experts, public manager players informed the winegrowers that they could not fund all demands for subsidies because this would represent about 600,000 €. Most of the cost of subsidies would result from winegrowers demands to change their sprayers (FT1-2, FT2A and FT2B) in the second round of game. Instead of funding a confining and recycling sprayer for each farm, they proposed to fund it in the framework of CUAM (Cooperative for the Use of Agricultural Material). In terms of grassing, they advocated for 1 inter-row in 2 instead of 1 in 4. However, following a discussion between winegrower and public manager players, only players of FT1-2 accepted to further modify their practices in the 2nd round and to choose Reasoned + for fungi management (they chose 0 CMR in the 1st round).

In the group of local actors W#2B1, the public manager players were surprised at the low number of subsidies requested for Agri-Environmental Measures (AEMs, e.g., grassing, Organic). They emphasized that the only subsidy requested in the 2nd round (confining sprayers for all FT1-2 farms) could not be funded because it would cost twice the watershed budget. The discussion between winegrower and

**Table 7**
Means-based results at watershed scale for the 1st round of the three CAPIPP game sessions for both years (as a % of total vineyard area). Note that FT2A was not played in W#2B1 and W#2B2.

| | High-performance sprayer | | Disease-resistant cultivars (%) | 0 pesticide on rows (%) | Organic (%) | Grassed headland (%) |
|---|---|---|---|---|---|---|
| | Number | Area (ha) | | | | |
| Initial | 0 | 0 | 0 | 18.2 | 18.2 | 11.6 |
| W#2A | 0 | 0 | 3.1 | 74.6 | 56.5 | 100 |
| W#2B1 | 0 | 0 | 0 | 18.2 | 18.2 | 93.4 |
| W#2B2 | 0 | 0 | 25.4 | 18.2 | 18.2 | 93.4 |

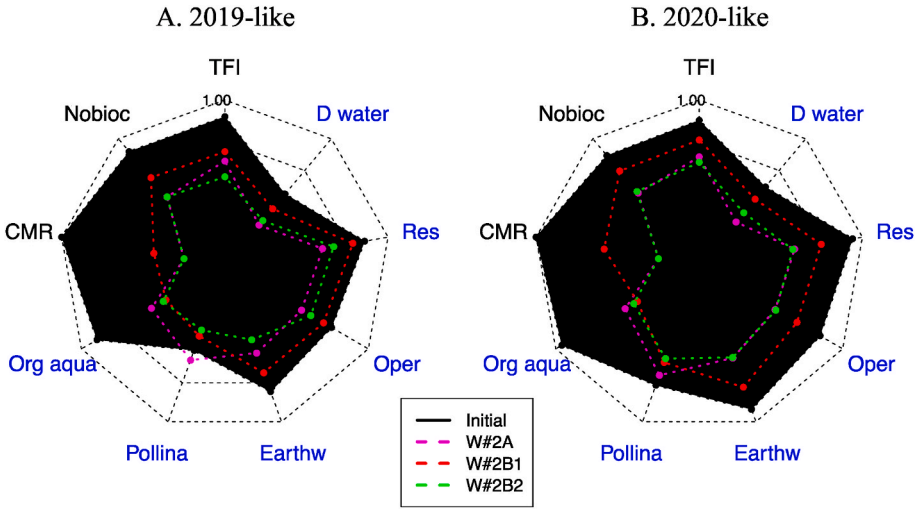



**Fig. 5.** Watershed evaluations for the three sessions. Labels of means-based indicators are in black text (TFI – Treatment Frequency Index; Nobioc – TFI without biocontrol products; CMR – Carcogenic Mutagen and Reprotoxic products), and labels of impact-based indicators are in blue text (Org aqua – Aquatic organisms; Pollina – Pollinators; Earthw – Earthworms; Oper – Operators; Res – Residents; D water – Drinking water catchment). For each indicator, minimum and maximum spider values correspond to the minimum and maximum value when considering all combinations of current and alternative practices.

public manager players highlighted that winegrowers preferred subsidies for material because they felt requirements for AEMs were not flexible enough (5-year commitment). Following this discussion, the player of FT1-2 decided he could not change his sprayer if no subsidy was available. No other players accepted to further modify their practices.

In the group of local actors W#2B2, the public manager player noted that very few winegrowers asked for subsidies. He proposed a subsidy for all winegrower players to change their sprayer for one that can decrease drift. The three winegrower players accepted, in the 2nd round, to change to air-blast sprayers, because confining and recycling sprayers are too costly and too constraining (spraying on only one row instead of two).

### 3.6. Debriefing

In all sessions, participants acknowledged the pedagogical value of the CAPIPP game because it "can initiate discussions and thoughts on strategies for change" and provides "an immediate visualization of changes in environmental potential impacts." They also acknowledged the conviviality and the richness of the exchanges between participants. All W#2B participants agreed, at least partially, on the realism of the changes and their applicability for other cultivars. Most of them found that only some changes were possible without subsidies. Only some of them were favorable to the purchase of sensitive fields by local authorities. They stressed that "it depends on the restrictions that are put on these fields." Among the alternatives that were not chosen, grassing, organic, 0 glyphosate, and confining sprayers were discussed (Appendix E). For the three first, yield and feasibility (including the issue of building a separate winemaking chain at the cooperative winery for organic grape production), were highlighted as critical. For confining sprayers, the additional working time, costs, and feasibility were seen as important limiting factors.

Finally, local players insisted on the difficulty of drastically changing their practices and offered several reasons: (1) the large areas to be managed and fields designed for mechanization, (2) the lack of flexibility in specifications (Organic, AEM for grassing) which made some winegrowers hesitate for fear of being penalized in unfavorable years, and (3) the questionable feasibility of disease-resistant cultivars in regard to their commercialization and inclusion in PDO labels. Among the propositions to overcome some of these issues, including labor and vulnerable fields in particular, were higher subsidies, which could be tied to technical support for winegrowers in their evolving practices.

Other limits, possible drawbacks, and solutions were also suggested. The absence of economic elements was underlined with a suggestion to increase financing capacity after transitioning to organic practices. Experts suggested recognizing different types of organic practices that involve different types of pesticide use. The difficulty of finding collective levers was highlighted. In the game, GDHO and CUAM are the only collective tools. Experts suggested additional levers, such as shared employees. Local actors suggested using artifacts (e.g., objects like mini sprayer) during the game. They felt there was too much information and that artifacts might give a tangible form to decisions and help structure the most important information to be considered (Appendix E).

## 4. Discussion

### 4.1. Advantages and limits of the method

According to the classification of Hassenforder et al. (2020), the serious game we designed had three principle aims: (1) enhancing knowledge transfer and learning (panels, spatial representation, etc.), (2) exploring new individual strategies (alternative choice), and (3) promoting relationships between players (exchanging viewpoints). With respect to learning, the experts made more use of information provided (panels, details on active substances). All groups used the spatial representation, and participants from all groups questioned the management alternatives and the assessment results. However, the local

participants suggested there was too numerous information and a lack of artifacts to make that information more accessible. Regarding artifacts, we could use something similar to Rouault et al., 2020, such as cards to represent vineyard operations. One participant suggested having access to the information in advance. Another way could be to dedicate a specific time for knowledge transfer. Thus, we only partially met the objective of knowledge transfer and learning. On the contrary, the objectives of exploring new individual strategies and promoting relationships between players, were clearly met.

Authors often cite the difficulties of maintaining actor involvement during workshops (Della Rossa et al., 2022). In our study, the initial assessment step was key to the winegrowers' involvement, and it launched the participatory process. This allowed us to involve more farmer profiles than in previous studies (Naulleau et al., 2022). However, we probably failed to involve "the most intensive" pesticide-users, since those actors tend to avoid these types of approaches.

First, we decided to include two roles in the game: farmers and a policy maker/watershed advisor, as seen in other games used to design agricultural systems for solving an environmental issue (e.g., herbicide pollution in Della Rossa et al., 2022; runoff in Souchère et al., 2010). Della Rossa et al. (2022) also involved technical advisors as players. According to our hypothesis on the divergence between strategies designed by experts compared to those of local players, we decided not to add such a role. However, future game sessions with local stakeholders could involve advisors to foster learning and stimulate more ambitious changes. Second, when playing the game we chose to assign, when possible, a role similar to the participant's real-life experience. This choice is common in role-playing and serious games (e.g., Barnaud et al., 2008; Castella et al., 2005; Della Rossa et al., 2022; Souchère et al., 2010). Other choices were possible: assigning preferred and then opposite roles (e.g., Etienne, 2003) or swapping the role between farmer and manager (e.g., Moreau et al., 2019). Those types of alternating roles could foster social learning (Villamor and van Noordwijk, 2011), but they are particularly useful in conflictual situations (e.g., Moreau et al., 2019). Assigning participants a familiar role allows them to be closer to a simulation of reality (Villamor and van Noordwijk, 2011) and avoids cases where players experience difficulties in assuming their role (e.g., Mathevet et al., 2007).

To build the game, it was necessary to simplify the reality by making several choices. Like Della Rossa et al. (2022), we grouped the farms into farm types in order to meet two objectives: (1) keep a realistic number of players that (2) will fully represent the main systems present in the watershed. To build a typology of these systems, we used a common statistical method for scaling up from farms to the territory (e.g., Chopin et al., 2015; Delmotte et al., 2017). The typology was then validated by the farmers. We also had to choose the description of the current pesticide uses for each farm type. For herbicides we observed low between-group variability in active substances and doses (as in Della Rossa et al., 2022), but fungicides at farm-type scale were more variable, both in their use and in their potential impacts. To maintain realism, we opted for the fungicide management of one surveyed farmer, whose potential impacts were above the median of the group. Another option would have been to propose that each farmer-player choose their initial practices, but this would have changed the initial situation. We did not choose this option because it would have complicated the game for the players.

In the CAPIPP calculations, we opted for simple potential indicators of impacts and avoided complicated indicators such as those predicting water quality, which depends on variables like watershed attributes, management practices, and hydrological factors (Khan et al., 2021). Other options were possible, for example, meta-modelling of complex models (e.g., Hossard et al., 2015), the hydrological model MHYDAS that simulates pollutant runoff at watershed scale (Biarnès et al., 2017), or the Pesticide Impact Rating Index (PIRI, Kookana et al., 2005). Those models, however, focus on water only and require time-consuming parameterization and simulations (MHYDAS) or information that is difficult to access (PIRI).

One last important limitation concerns the economic and yield impacts that were not provided in the strategy assessments. This issue was identified by one of the experts: "not enough about the economic aspects, which is what winegrowers want to hear about." This assessment is critical with regard to risk management (e.g., Lescot et al., 2014), but the impacts of pests and diseases on vine yield and wine quality remain difficult to predict, especially with regard to fungi management, and they require large datasets for model parameterization (e.g., Leroy et al., 2013).

### 4.2. Designed strategies

Coming back to our hypothesis regarding the different groups, we noticed differences on several aspects: experts requested more subsidies (inter-row grassing, anti-drift sprayer) and were more ambitious on fungi management and glyphosate use. Regarding subsidies, we saw a clear difference in perceptions: while experts qualified them as "helping a transition," local actors underlined their "inflexibility" (5-years engagement for a practice, e.g., row grassing), and "low economic value." Similarly, most local actors found the anti-drift sprayers "too costly." The elimination of glyphosate use and changes in fungi management could be linked to differences in knowledge, field-practices, and risk perception of both pesticides impacts and possible yield losses. Experts are familiar with decision tools for pest management whereas farmers are not. Farmers are much more sensitive to management risks than experts. For example, winegrowers "live the difficulty of managing weeds," while experts found it "manageable." Like Teschner and Orenstein (2021), in terms of risk, we observed that winegrowers saw changing fungi strategy as economically dangerous. Adding information on risk (e.g., quantifying the impact on economic performance from changes in practices) could help to reduce risk perception (Perrin et al., 2022). An extreme choice was performed in both local and expert groups: moving almost all the farm area to organic for FT1-1 (experts), and to disease-resistant cultivars (local, W#2B). However, these choices were selected by winegrowers in cooperative wineries, who have little independence to make such strategic choices (Teschner et al., 2017). Interestingly, extremes were not chosen in the group that included the head of the local winery cooperative. He explained his reservations about an organic choice: "It requires an investment that is high … at the level of the cooperative, because it requires different ways, a dubbing of docks. For me, today, why don't I go there? Because it is difficult." Conversely, cooperatives can act as a lever for stimulating change in practices (Teschner and Orenstein, 2021). Training requests were numerous in all groups and on various themes. Training was also highlighted by Della Rossa et al. (2022) as an innovation that can foster change.

The collective insect management tool was promoted by all. However, it has to be managed by winegrowers, and the only tool like this that exists is in the Northern part of the area. Attempts have been made to establish one in the Southern area, but they were abandoned due to lack of funding. Groups for sharing experiences were also proposed. Similar aid tools were also proposed by stakeholders in Della Rossa et al. (2022). The important questions remain on how to organize these types of tools and make them sustainable.

Interestingly, all participants were more willing to make efforts on the block next to houses (1 ha). We could ask if this would be true for larger areas, and for other vulnerable areas, such as next to watercourses. Grimene et al. (2021) proposed a method for spatializing vulnerable areas with regards to water pollution. This spatial information could be added to the game, with a new block, to design specific strategies targeting water quality.

Overall, the changes that players selected led to the decrease in both the use of pesticides (means-oriented indicators) and the potential environmental impacts (results-based indicators). Despite two exceptions (one choosing organic, and another choosing disease-resistant

varieties), efforts made by the players, especially in winegrower sessions, remained limited to punctual actions and did not completely modify the farming system. The players justified their preference for moderate changes, for instance for fungi or weed management, by pointing to the extra costs of the related increase in labor and monitoring. Although Perrin et al. (2022), in a co-design workshop, showed that it was possible to decrease impacts from farming pesticide use while increasing economic performance, our results are consistent with the findings of Beckmann and Wesseler (2003). Nevertheless, even limited changes led to decreases in the values of indicators at both farm and watershed scales. However, larger decreases could have been obtained if more radical changes were widely chosen (i.e., "0" value in Fig. 5). The only exception was the use of CMR active substances, which was strongly reduced in two out of the three game sessions. These limited decreases in pesticide use and impacts in vineyards are consistent with the results of Perrin et al. (2022). Using a plot-scale serious game in eco-design workshops with winegrowers, they showed a reduction of only 4% in the aggregated environmental impact. Methods to encourage alternative practices have been proposed in other studies. For example, Della Rossa et al. (2022) employed the Knowledge-Concept method proposed by Hatchuel and Weil (2009) to help actors think in a disruptive way. Perrin et al. (2022) worked in groups at the scale of an actual farm, where participants encouraged the farmer to adopt more environmental-friendly practices.

Finally, comprehensive changes are conditioned by socio-technical lock-ins and path dependency (Cowan and Gunby, 1996), which were identified by Guichard et al. (2017) as a source of the failure of French plans to reduce pesticide use. Indeed, farmers are entrenched in pesticide-dependent systems with strong interdependencies among economic actors (Guichard et al., 2017) that hinder the emergence of innovative agricultural systems.

## 5. Conclusion

We developed and presented a serious game to stimulate thinking about alternative strategies of pesticide use in vineyards. One advantageous and less common aspect of our approach is the targeting of all types of pesticides and a large range of potentially impacted elements (e.g., earthworms, water quality, residents). Another original aspect can be seen in the multi-scale perspective we adopted for both strategy design and assessment: blocks, farm, and watershed. And finally, we enriched this perspective by considering combinations of levers and incorporating a diversity of choices. Our approach, described as "pedagogical" by players, could be adapted to other regions, assuming a spatial analysis of the existing situation. Even without adaptations to a specific new watershed, it can be used to consider possible changes to wine-growing practices. Associating those changes to an economic analysis would represent an improvement, although such an analysis would be difficult to perform due to lack of knowledge on the relationships between pest and disease pressure, pesticide application, vine yield, and wine quality. The different attitudes between local actors and experts also revealed different perceptions on investment, risks, and subsidies. Sharing the results with policymakers could help highlight the gap between policy targets and their application under real-world conditions, emphasizing the advantages of introducing flexibility in the conditions for subsidies.

## CRediT authorship contribution statement

**L. Hossard:** Conceptualization, and, Methodology, Investigation, Writing – original draft, Funding acquisition. **C. Schneider:** Conceptualization, and, Methodology, Investigation, Writing – review & editing. **M. Voltz:** Conceptualization, and, Methodology, Investigation, Writing – review & editing, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclepro.2022.134913.

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